

Physics Formulas

Energy required to separate ions:

$$W = \Delta K + \Delta U = 0 - U = -\frac{kq_1q_2}{r} = \frac{ke^2}{r}$$

Converting W to eV:

$$W_{ev} = \frac{W}{1.6e-19}$$

Capacitance of Capacitor:

$$C = \frac{k\epsilon_0 A}{d}$$

$$C = \frac{Q}{V}$$

$$C_{new} = kC_{old}$$

$$U_{new} = \frac{q^2}{2C_{new}}$$

Energy in Capacitor:

$$E = \frac{QV}{2} = \frac{CV^2}{2}$$

Flux through surface:

$$\Phi = \frac{q}{\epsilon_0}$$

Flux through cube:

$$\Phi = \frac{q}{6\epsilon_0}$$

Maximizing force of repulsion of charges q

and Q - q:

$$\frac{dF}{dq} = \frac{k}{r^2}(Q - 2q) = 0$$

Electric Dipole Moment:

$$P = qr$$

EF from Point Charge:

$$\frac{kQ}{r^2}$$

EF from line charge of finite length:

$$E = E_x + E_y$$

$$E_x = k\lambda\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

EF far from finite line charge:

$$E = \frac{kQ}{r^2}$$

EF from infinite line charge:

$$\frac{2k\lambda}{R}$$

EF on axis of charged ring:

$$\frac{kQz}{(z^2 + a^2)^{\frac{3}{2}}} = \frac{kQz}{r^3}$$

EF on axis of charged disk:

$$\frac{kQ}{z^2}$$

EF from two infinite planes:

From one: $\frac{\sigma}{2\epsilon_0}$

From both: $E_1 + E_2$

Flux through piecewise-continuous closed surface:

$$\Phi = \Phi_{right} + \Phi_{left} + \Phi_{curved}$$

$$\Phi_{curved} = 0$$

$$\Phi_{right} = EA$$

EF from uniformly charged slab:

$\frac{\rho a}{\epsilon_0}$ where p is charge density and a is half the slab length

$$E = \frac{\sigma}{2\epsilon_0}$$

EF from thin spherical shell:

Inside $E = 0$

Outside $E = \frac{kQ}{r^2}$ where $Q = \sigma 4\pi R^2$

EF from uniformly charged solid sphere of radius R:

$$r > R \quad E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$

$$r < R \quad E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} r$$

Electrical Potential of single charge:

$$V = \frac{kQ}{r}$$

Electric potential energy of system:

$$U = \frac{kq_1q_2}{r} = \frac{QV}{2} = \frac{CV^2}{2} = \frac{Q^2}{2C}$$

Converting electric potential to electric potential energy:

$$U = qV$$

Work done in moving charge from A to B:

$$\Delta V = V_B - V_A$$

Potential from dipole:

$$V = \frac{kP}{r^2}$$

Potential on axis of disk:

$$V = 2\pi k\sigma|x|\left(\sqrt{1 + \frac{R^2}{z^2}} - 1\right)$$

Potential at distance R from point charge and sheet of charge:

$$V = -2\pi k\sigma|x| + kq\left(\frac{1}{r} - \frac{1}{a}\right)$$

Relationship between E and V:

$$E = -\nabla V$$

Torque exerted by field on dipole:

$$\tau = pE\sin(\theta)$$

EF between 2 curved rods:

$$\frac{4kq}{\pi r^2}$$

Charge between 2 concentric spheres:

$$Q_B = q_o - (q_A - q_i) - q_i$$

Gauss' Law for a cylinder:

$$2E\pi rL = \frac{q}{\epsilon_0}$$

where $q = p\pi(r_o^2 - r_i^2)L$

EF from infinitely long cylindrical wire:

$$\text{Outside: } E = \frac{\sigma r}{\epsilon_0 x}$$

Geometrical Formulas

Sphere

Surface Area: $4\pi r^2$

Volume: $\frac{4}{3}\pi r^3$

Cylinder

Surface Area: $2\pi r^2 + 2\pi rh$

Volume: $\pi r^2 h$

Cube

Surface Area: $6a^2$

Volume: s^3

Tips

On changing capacitance:

Capacitance changed while connected, charge changes.

Capacitance changed while disconnected, voltage changes.

If reconnected, voltage readjusts and energy will change.